Black-Scholes Option Pricing Model

Introduction

Derivative securities are financial instruments whose value is derived from an underlying asset such as a stock, commodity, or financial index. One of the most common derivatives is an **option**, which gives the holder the right, but not the obligation, to buy or sell an asset at a predetermined price within a specified time period. A **call option** gives the right to **buy** the asset at the strike price, while a **put option** gives the right to **sell** the asset at the strike price.

Accurate option pricing is important in financial markets for trading, risk management, and investment decisions. The **Black–Scholes model** is one of the most widely used frameworks for pricing European options. The model provides a closed-form analytical formula that determines the theoretical value of an option based on factors such as the **stock price, strike price, volatility, risk-free interest rate, and time to maturity**. It assumes that stock prices follow a lognormal distribution and evolve continuously over time. In addition, **Monte Carlo simulations** can be used to model possible future stock price paths and estimate option values.

The objective of this project is to estimate the theoretical price of a European call option on **Reliance Industries** using the **Black–Scholes model** and validate the result using **Monte Carlo simulation**. The project also aims to analyse the sensitivity of the option price to key market variables through the calculation of the **option Greeks**.

Data and Asset Description

The analysis in this project is based on **Reliance Industries Ltd.**, one of the largest publicly traded companies in India and a major constituent of the **Indian equity market**. The company was selected as the underlying asset due to its high liquidity, large market capitalization, and the availability of reliable historical trading data.

Asset Selection and Data Source

- **Reliance Industries Ltd.** was chosen as the underlying asset for the option pricing analysis.

- Historical stock price data was obtained from the **National Stock Exchange of India (NSE).**

Time Horizon

- The dataset consists of **three years of daily stock price data**, covering the period from **January 2023 to February 2026**.

- This time period provides a sufficiently large dataset for analysing stock return behaviour and estimating volatility.

The data was imported into **R** and organised into a continuous time series of **daily stock prices**. Stock price values were converted into **numeric format** to enable statistical analysis.

Historical price data is primarily used to estimate the **volatility of the underlying stock**, which is one of the most important inputs in the **Black–Scholes option pricing model**. Volatility measures the **degree of fluctuation in stock prices over time** and reflects the uncertainty or risk associated with the asset. Since option values are highly sensitive to changes in volatility, accurately estimating this parameter is crucial for obtaining reliable option pricing results.

Return Construction and Statistical Overview

Log Return Calculation

Stock returns were calculated using **logarithmic returns**, defined as:

$r_t = \ln (P_t / P_{t-1})$

where P_t represents the stock price at time t and P_{t-1} represents the price on the previous trading day.

Log returns are commonly used in **financial modelling** because they are **time-additive**, easier to analyse statistically, and consistent with the assumption that stock prices follow a **lognormal distribution**, which is a key assumption in the **Black–Scholes model**.

Summary Statistics

The distribution of the calculated **daily log returns** was examined using summary statistics, including the **mean**, **median**, **quartiles**, and the **minimum and maximum values**. The computed statistics are summarised below:

- **Minimum:** -0.6882640

- **1st Quartile:** -0.0076981
- **Median:** 0.0005907
- **Mean:** -0.0006985
- **3rd Quartile:** 0.0070150
- **Maximum:** 0.0678380

These statistics provide an overview of the central tendency and spread of the return distribution. The results indicate that most daily returns are relatively small and clustered around zero, while occasional larger positive or negative returns occur. This behaviour is typical for financial time series and reflects the **volatile nature of stock price movements**.

Trading Day Verification

The number of trading days observed in the dataset was examined for each year. The results show that a typical full year contains approximately **250 trading days**.

In financial modelling, it is standard practice to assume **252 trading days per year** when converting **daily volatility into annualised volatility**. This convention is widely used in quantitative finance and provides a consistent basis for volatility estimation and option pricing calculations.

Volatility Estimation

Volatility is a key parameter in option pricing, as it measures the degree of variability in the price of the underlying asset. In the **Black–Scholes model**, volatility represents the expected fluctuation of stock prices over time and has a significant impact on the value of an option.

Daily Volatility

Daily volatility was estimated using the **standard deviation of the daily log returns**. The standard deviation measures the dispersion of returns around their mean and provides an estimate of the typical daily movement in the stock price.

The calculated **daily volatility** of Reliance Industries stock was:

$\sigma_{daily} = 0.02847035$

Annualized Volatility

Since option pricing models typically require **annualized volatility**, the daily volatility was converted using the **square-root-of-time rule**:

$\sigma_{annual} = \sigma_{daily} * \sqrt{252}$

where **252** represents the standard number of trading days in a year.

The resulting **annualised volatility** was:

$\sigma = 0.4519527$

Interpretation

The estimated **annualised volatility of approximately 45.2%** indicates that the price of Reliance Industries stock exhibits a moderate to high level of variability over time. In the context of option pricing, higher volatility increases the potential range of future stock prices and therefore raises the value of options, as it increases the likelihood that the option will finish **in-the-money**. Consequently, accurate estimation of volatility is crucial for obtaining reliable option pricing results using the **Black–Scholes framework**.

Black–Scholes Option Pricing Framework

The **Black–Scholes model**, developed by **Fischer Black and Myron Scholes (1973)**, is one of the most widely used models for pricing **European-style options**. The model provides a **closed-form analytical formula** to determine the theoretical value of an option based on key market variables such as the **stock price**, **strike price**, **volatility**, **risk-free interest rate**, and **time to maturity**.

Model Assumptions

The Black–Scholes framework is based on several simplifying assumptions about financial markets:

- **Lognormal stock price dynamics:** Stock prices follow a stochastic process that results in **lognormally distributed prices**.

- **Constant volatility:** The volatility of the underlying asset remains constant over the life of the option.

- **Frictionless markets:** Markets operate without **transaction costs**, **taxes**, or other trading frictions.
- **No arbitrage:** There are no opportunities to make **risk-free profits** through arbitrage.
- **Continuous trading:** Investors can trade continuously and adjust their positions at any time.

These assumptions allow the derivation of a mathematical formula for pricing European options.

Black–Scholes Call Option Formula

The theoretical price of a **European call option** is given by:

$C = S_0 N(d_1) - Ke^{-rT} N(d_2)$

where:

- C = price of the **European call option**
- S_0 = **current stock price**
- K = **strike price**
- r = **risk-free interest rate**
- T = **time to maturity** (in years)
- N(.) = **cumulative distribution function of the standard normal distribution**

The intermediate terms d_1 and d_2 are defined as:

$d_1 = (\ln (S_0 / K) + (r + (½) * \sigma^2) * T) / (\sigma\sqrt{T})$

$d_2 = d_1 - (\sigma\sqrt{T})$

where σ represents the **volatility of the underlying asset**.

The variables d_1 and d_2 incorporate the effects of **time to maturity, volatility, and the risk-free interest rate** in determining the option price. Intuitively, $\mathbf{d_1}$ can be interpreted as a risk-adjusted measure of how far the current stock price is from the strike price, while $\mathbf{d_2}$ reflects the probability that the option will finish in-the-money at maturity under the risk-neutral framework.

Model Inputs

To implement the **Black–Scholes option pricing model**, five key parameters must be specified. These inputs determine the theoretical value of the option and represent the prevailing market conditions of the underlying asset.

Current Stock Price (S_0)

The **current stock price** represents the market value of the underlying asset at the time of valuation. In this analysis, the latest **closing price of Reliance Industries** from the dataset was used as the current stock price.

Strike Price (K)

The **strike price** is the price at which the option holder has the right to purchase the underlying asset. In this analysis, the strike price was set equal to the current stock price, creating an **at-the-money (ATM) option**, where the **stock price and strike price are approximately equal**. ATM options are commonly used as a benchmark in option pricing analysis.

Risk-Free Interest Rate (r)

The **risk-free interest rate** represents the return on an investment with no risk of default. In this project, a risk-free rate of **6.7%** was assumed, based on the yield of **Indian government securities**, which are generally considered risk-free.

Volatility (σ)

The **volatility** parameter represents the expected variability in the price of the underlying stock. In this study, volatility was estimated from the **historical log returns** of Reliance Industries and annualized using the **square-root-of-time rule**. The resulting annualized volatility was **0.4519527 (45.2%)**.

Time to Maturity (T)

The **time to maturity** represents the remaining life of the option, expressed in years. In this analysis, a maturity of **30 days** was assumed, corresponding to:

$T = 30/365$

which represents a **one-month option contract**.

Summary of Model Inputs

Parameter	Description	Value
S_0	Current stock price	1393.9
K	Strike price	1393.9
r	Risk-free interest rate	0.067
σ	Annualized volatility	0.4519527
T	Time to maturity	30/365

These parameters are used as inputs in the **Black–Scholes formula** to compute the theoretical value of the **European call option**.

Analytical Option Pricing Results

Using the specified model inputs, the intermediate Black–Scholes variables were calculated as:

$d_1 = 0.1073$

$d_2 = -0.0223$

Substituting these values into the **Black–Scholes call option pricing formula** gave the theoretical value of the **European call option** as:

$C = 75.70$

Thus, the estimated fair price of a **30-day at-the-money European call option** on **Reliance Industries** is approximately **₹75.70**.

This result represents the price an investor would pay today for the right to purchase the stock at the strike price on the maturity date. Since the option is **at-the-money**, its value is derived mainly from its **time value** and the expected **future volatility** of the underlying stock.

Monte Carlo Simulation Framework

Geometric Brownian Motion

In order to simulate future stock prices, the underlying asset price is assumed to follow a **Geometric Brownian Motion (GBM)** process. GBM is widely used in financial modelling because it captures the random and continuous nature of stock price movements while ensuring that prices remain positive.

The GBM model describes the evolution of stock prices as:

$dS_t = \mu S_t dt + \sigma S_t dW_t$

where:

- S_t is the **stock price at time** t
- μ represents the **drift** or expected return
- σ represents the **volatility of the stock**
- dW_t is a **Wiener process (Brownian motion)** representing random market shocks

Under this framework, stock prices follow a **lognormal distribution**, which is consistent with the assumptions of the **Black–Scholes model**.

Simulation Procedure

Using the estimated parameters, **Monte Carlo simulation** was applied to generate possible future stock prices. The simulation involves generating random values that represent potential market movements and using them to simulate the evolution of stock prices over time.

A total of **10,000 simulated stock price paths** were generated to model possible values of the stock price at the option's maturity.

Distribution of Simulated Prices

The simulation produces a **distribution of possible future stock prices** rather than a single predicted value. This distribution reflects the uncertainty in future price movements and captures a wide range of potential outcomes.

Most simulated prices tend to cluster around the **current stock price**, while some scenarios produce significantly higher or lower values. This range of outcomes is later used to compute **option payoffs** and estimate the **Monte Carlo option price**.

Monte Carlo Option Pricing

After generating simulated stock prices using **Monte Carlo simulation**, the payoff of a **European call option** was computed for each simulated outcome.

Payoff Function

The payoff of a European call option at maturity is defined as:

Payoff = max (S_T - K, 0)

where:

- S_T represents the **simulated stock price at maturity**
- K represents the **strike price**

This payoff structure ensures that the option is exercised only when the stock price exceeds the strike price. If the simulated price is lower than the strike price, the payoff is zero.

Discounting the Expected Payoff

Once the payoff was calculated for all **10,000 simulated prices**, the **average payoff** was computed. This expected payoff was then **discounted at the risk-free interest rate** to obtain the present value of the option price:

$C = e^{-rT} E[\max (S_T - K, 0)]$

where r is the **risk-free interest rate** and T is the **time to maturity**.

The **Monte Carlo estimate of the call option price** was calculated as:

C_{MC} = 75.83863

Comparison with Black–Scholes Price

The Monte Carlo estimated option price of **₹75.84** is very close to the **Black–Scholes price of ₹75.70** obtained earlier. The close agreement between these two values indicates that the simulation results are consistent with the analytical pricing model, confirming the validity of the pricing framework under the assumption that the stock price follows **Geometric Brownian Motion**.

Sensitivity Analysis: Option Greeks

In option pricing, **Greeks** measure the sensitivity of an option's value to changes in key market variables. These measures are widely used in **risk management and trading strategies** to understand how option prices respond to movements in the underlying asset, volatility, and time.

Delta (Δ)

Delta measures the sensitivity of the option price to changes in the **underlying stock price**.

Δ = 0.542719

A Delta value of **0.5427** indicates that a **₹1 increase in the stock price** would lead to an approximate **₹0.54 increase in the option price**, holding other factors constant.

Gamma (Γ)

Gamma measures the **rate of change of Delta** with respect to changes in the stock price.

Γ = 0.0021962

A positive Gamma indicates that **Delta increases as the stock price rises**, reflecting the **convex relationship between option value and stock price**.

Vega

Vega measures the sensitivity of the option price to changes in **volatility**.

Vega = 158.5099

A relatively high Vega indicates that the option value is **highly sensitive to changes in volatility**, which is typical for **at-the-money options**.

Theta (Θ)

Theta measures the effect of **time decay** on the option price.

Θ_{annual} = -481.4174

Θ_{daily} = -1.318952

The negative Theta value indicates that the option **loses value as time passes**, assuming other factors remain constant. On a daily basis, the option price decreases by approximately **₹1.32 due to time decay**.

Interpretation

The computed Greeks suggest that the option exhibits **moderate sensitivity to changes in the underlying stock price (Delta)**, **positive convexity (Gamma)**, **strong responsiveness to volatility changes (Vega)**, and **negative time decay (Theta)**. These characteristics are consistent with the behaviour of a **short-term at-the-money European call option**.

Simulation Visualization and Interpretation

To better understand the results of the **Monte Carlo simulation**, histograms were generated for both the **simulated stock prices at maturity** and the resulting **option payoffs**.

Distribution of Simulated Stock Prices

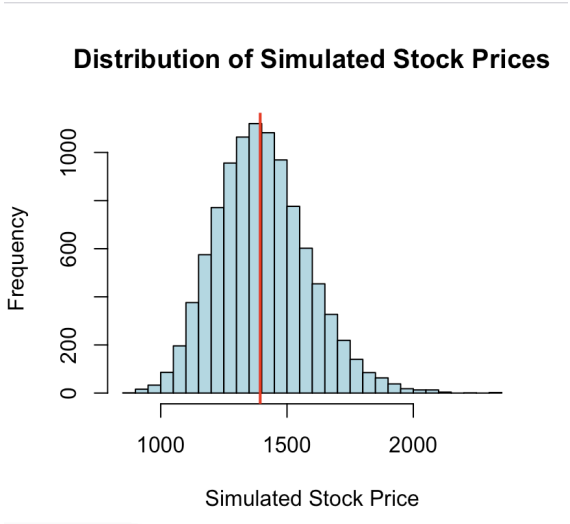

Figure 1: Distribution of Simulated Stock Prices

The first histogram shows the **distribution of simulated stock prices** at the option's maturity. The red vertical line represents the **current stock price of Reliance Industries (≈ ₹1393.9)**.

The distribution is centered close to the current stock price, indicating a strong **central tendency** around the present market value. Most simulated prices lie roughly between **₹1200 and ₹1700**, suggesting that the most probable future outcomes remain near the current price level.

The histogram also exhibits a **right-skewed shape**, meaning that while stock prices have a limited downside (they cannot fall below zero), there is significant potential for higher prices. The **right tail** of the distribution represents scenarios where the stock price rises substantially.

This pattern is consistent with the **lognormal distribution** implied by the **Black–Scholes framework**, where stock prices are assumed to evolve according to **Geometric Brownian Motion**.

Distribution of Option Payoffs

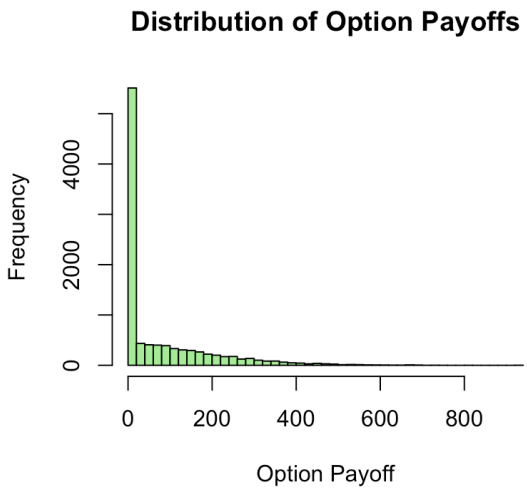

Figure 2: Distribution of Simulated Call Option Payoff

The second histogram illustrates the **distribution of call option payoffs** obtained from the simulated stock prices. A large number of outcomes result in **zero payoff**, which occurs when the simulated stock price is **below the strike price** at maturity and the option expires **out-of-the-money**.

For scenarios where the stock price exceeds the strike price, the payoff increases as the difference between the **stock price and strike price** becomes larger. This results in a **right-skewed payoff distribution**, with a small number of scenarios generating very high payoffs.

Together, these visualizations demonstrate how **Monte Carlo simulation produces a range of possible future outcomes** for both the underlying asset and the resulting option payoff, providing a probabilistic perspective on option valuation

Model Assumptions and Limitations

The **Black–Scholes model** and the **Monte Carlo simulation framework** rely on several simplifying assumptions about financial markets. These assumptions allow the development of mathematical pricing models but may not fully reflect real market conditions.

Model Assumptions

The key assumptions underlying the modelling framework are:

- **Constant volatility:** The volatility of the underlying asset remains constant throughout the life of the option.
- **Lognormal price dynamics:** Stock prices follow **Geometric Brownian Motion**, resulting in **lognormally distributed prices**.
- **No transaction costs:** Markets are assumed to be **frictionless**, meaning there are no transaction costs, taxes, or liquidity constraints.
- **Constant risk-free interest rate:** The risk-free interest rate remains fixed over the life of the option.
- **Continuous trading:** Investors can trade continuously and adjust their positions at any time.
- **No arbitrage opportunities:** Markets operate efficiently without opportunities for **risk-free profits**.

Limitations

In practice, financial markets may deviate from these assumptions:

- **Volatility is not constant** and can change significantly due to market conditions or economic events.

- **Stock prices may experience sudden jumps or extreme movements** that are not fully captured by Geometric Brownian Motion.
- **Transaction costs and bid–ask spreads** exist in real markets and can affect trading decisions and pricing.
- **Interest rates may vary over time**, whereas the model assumes a constant risk-free rate.
- **Market frictions and liquidity constraints** may limit the ability to trade continuously.

Therefore, the option prices obtained from the model should be interpreted as **approximations under simplified market conditions**, rather than exact representations of real market prices.

Conclusion

This project applied the **Black–Scholes option pricing model** and **Monte Carlo simulation** to estimate the value of a **European call option** on **Reliance Industries** using historical stock price data.

The analysis began with the calculation of **logarithmic returns** and the estimation of **stock volatility**, which was found to be approximately **45.2% on an annualized basis**. Volatility plays a critical role in option pricing, as it determines the extent of possible fluctuations in the underlying asset price.

Using the estimated parameters, the **Black–Scholes model** produced a call option price of approximately **₹75.70** for a **30-day at-the-money option**. To validate this analytical result, a **Monte Carlo simulation** with **10,000 simulated price paths** was performed. The simulation produced an estimated option price of approximately **₹75.84**, which is very close to the Black–Scholes value, confirming the consistency of the pricing framework.

The analysis of **option Greeks** further provided insights into the sensitivity of the option price to changes in key variables such as the **underlying stock price, volatility, and time to maturity**. The results indicated moderate sensitivity to stock price movements, strong responsiveness to volatility changes, and the presence of time decay as the option approaches maturity.

Overall, this project demonstrates how quantitative financial models can be applied to estimate option prices and analyse risk in financial markets. Despite their simplifying assumptions, models such as the Black–Scholes framework and Monte Carlo simulation remain fundamental tools in quantitative finance for option valuation, risk management, and financial decision-making.